

Part A: Negative Number Operations*Evaluate each expression completely.*

1. $11 - 5 \cdot (2 + 11)$

15. $8 - 3 \cdot ((-9) + 8)$

2. $(-9)^3 - 15 \cdot 6$

16. $(-9 + 32)^2 \div 1$

3. $(-2)^2 - (-4) \cdot 3$

17. $(-4)^3 - 20 \cdot 11$

4. $(-8)^2 - (-12) \cdot 3$

18. $(-3)^3 - (-38) \cdot 9$

5. $(-3)^2 - 17 \cdot 4$

19. $3^2 - 3 \cdot 8$

6. $6 - 4 \cdot ((-5) + 6)$

20. $(-8)^2 - (-22) \cdot 11$

7. $(-7)^2 - 19 \cdot 11$

21. $(6 + 33)^2 \div 3$

8. $-14 - 4 \cdot (6 + (-14))$

22. $(9 + 0)^2 \div (-3)$

9. $11 - 10 \cdot (4 + 11)$

23. $(5 + (-13))^2 \div 2$

10. $(2 + 9)^2 \div (-1)$

24. $2^2 - 3 \cdot 10$

11. $(-3)^3 - 9 \cdot 6$

25. $(-5 + 8)^2 \div 3$

12. $-12 - 10 \cdot (5 + (-12))$

26. $(5 + 4)^2 \div 3$

13. $-15 - 11 \cdot ((-3) + (-15))$

27. $(-4 + 35)^2 \div 1$

14. $3^2 - (-6) \cdot 7$

28. $(-4)^2 - 40 \cdot 10$

29. $(-2)^3 - (-22) \cdot 3$

30. $4^2 - 4 \cdot 6$

Part B: Negative Number Operations*Evaluate each expression completely.*

1. $-9 - (-5)$

14. $12 \cdot (-11)$

2. $-5 - 3$

15. $-4 \cdot 11$

3. $-18 - (-7)$

16. $-169 \div (-13)$

4. $-14 + (-9)$

17. $-3 \cdot (-6)$

5. $-5 + (-4)$

18. $-9 \cdot (-7)$

6. $38 - (-5) - (-23)$

19. $-15 \cdot 4$

7. $33 - 37 - (-14)$

20. $56 \div (-4)$

8. $19 - 24 - 33$

21. $-6 \div (-2) - 14$

9. $-36 - (-3) - (-18)$

22. $-90 \div (-10) - 7$

10. $-22 - 31$

23. $-2 + (-2) \cdot 8$

11. $-42 \div (-6)$

24. $12 \div (-4) - 2$

12. $-11 \cdot 12$

25. $-3 + 8 \cdot 12$

13. $-90 \div (-9)$

26. $84 \div 12 - 40$

27. $-96 \div (-8) - 33$

29. $-17 + 13 \cdot (-10)$

28. $10 + (-10) \cdot (-8)$

30. $-30 + 14 \cdot (-3)$

Part C: Solving Linear Equations*Solve each equation for x . Show your steps.*

1. $x + 10 = 5$

9. $11x - 12 = -\frac{57}{2}$

2. $x + 9 = 6$

10. $8(2x + 12) + 3 = 259$

3. $12x = 72$

11. $6(5x + 7) - 4 = 8$

4. $2x = 20$

12. $2(2x - 8) - 10 = -38$

5. $7x + 3 = \frac{55}{2}$

13. $2(4x + 12) + 2 = -30$

6. $11x - 5 = 17$

14. $3(7x + 3) - 9 = -63$

7. $9x - 1 = 17$

15. $-4x + 16 = 15x + 149$

8. $2x + 12 = 2$

16. $-10x + 5 = 8x + 185$

17. $18x - 19 = 10x - \frac{25}{3}$

24. $-\frac{3}{2}x + 10 = 7$

18. $-6x + 4 = 2x - 28$

25. $x - 4 = -\frac{2}{5}$

19. $3x - 17 = -2x + 58$

26. $\frac{9x + 142}{4} = \frac{7}{4}$

20. $-6x - 9 = -3x + 33$

27. $\frac{3x - 61}{2} = \frac{7x - 145}{6}$

21. $\frac{x}{5} + 4 = \frac{34}{5}$

28. $\frac{3x - 26}{6} = \frac{-17}{6}$

22. $-\frac{5}{4}x - 3 = -8$

29. $\frac{-7}{x} = \frac{1}{2}$

23. $\frac{x}{6} + 12 = \frac{47}{4}$

30. $\frac{-4}{x} = \frac{6}{5}$

Part D: Linear Inequalities

Solve each inequality for x .

1. $x + 12 > 6$

3. $x - 9 \geq 4$

2. $x + 7 < -7$

4. $x - 4 \geq -11$

5. $-9x > -45$

15. $-7x + 19 > 19x - 397$

6. $11x + 3 \leq 113$

16. $-19x - 11 < 9x - 123$

7. $5x + 8 > -12$

17. $-5x - 19 > 3x - 75$

8. $-6x + 5 < -7$

18. $-7x - 9 > 3x + 111$

9. $5x - 1 > 59$

19. $\frac{x}{3} - 18 \geq -\frac{49}{3}$

10. $-2x - 11 \geq -27$

20. $-2x - 1 \geq -29$

11. $-x + 10 < 13x + 52$

21. $\frac{x}{5} - 12 < -11$

12. $2x - 18 \leq -17x - 246$

22. $\frac{x - 10}{2} + 2 \geq -\frac{11}{2}$

13. $-9x - 17 \geq 5x - 227$

23. $-\frac{3}{2}x - 14 \geq 16$

14. $5x + 1 < 17x - 47$

24. $\frac{3}{5}x + 18 > \frac{81}{5}$

25. $-x + 8 > \frac{3}{2}x - \frac{19}{2}$

28. $-\frac{3}{4}x + 9 > \frac{5}{2}x - \frac{3}{4}$

26. $\frac{5}{6}x - 15 \geq \frac{4}{3}x - 19$

29. $x + 20 < -\frac{5}{3}x + \frac{28}{3}$

27. $x - 1 < \frac{5}{4}x + \frac{1}{2}$

30. $\frac{9}{2}x + 11 < \frac{5}{3}x + \frac{101}{3}$

Part E: Slope and Linear Graphs

Identify the slope m and the y -intercept b of each line.

1. $y = \frac{7}{2}x + 6$

3. $y = 8x + 7$

2. $y = 6x + 9$

4. $y = 5x - 6$

5. $y = 9x + 1$

Part F: Slope Through Two Points

Find the slope of the line through each pair of points.

1. $(2, 1)$ and $(5, 7)$

4. $(-8, -21)$ and $(5, 18)$

2. $(2, 15)$ and $(8, 33)$

5. $(2, 1)$ and $(4, 2)$

3. $(4, 8)$ and $(8, 18)$

Part G: Equations From Two Points

Write the equation of the line through each pair of points in slope-intercept form.

1. $(-12, -4)$ and $(-2, 1)$

2. $(-12, -35)$ and $(4, 21)$

3. $(1, -6)$ and $(4, -6)$

5. $(-10, -23)$ and $(-8, -18)$

4. $(-4, -3)$ and $(4, 9)$

Part H: Equations From a Slope and a Point

Write the equation of each line in slope-intercept form.

1. $m = \frac{4}{3}$ through $(-6, -6)$

4. $m = \frac{1}{2}$ through $(-6, -3)$

2. $m = \frac{8}{3}$ through $(6, 16)$

5. $m = \frac{5}{3}$ through $(-6, -5)$

3. $m = \frac{9}{2}$ through $(0, 3)$

Part I: Slope and Equation From Two Points

Find the slope of the line through each pair of points, then write the equation of the line in slope-intercept form.

1. $(-3, -1)$ and $(0, 10)$

4. $(8, 23)$ and $(12, 32)$

2. $(-4, -7)$ and $(-3, -7)$

5. $(8, 11)$ and $(10, 16)$

3. $(-4, -10)$ and $(4, 14)$

Part J: Rewriting in Slope-Intercept Form

Rewrite each equation in slope-intercept form, then identify the slope m and the y -intercept b .

1. $14x - 4y = 4$

4. $-8x + 2y = 6$

2. $3y = 6x + 6$

5. $6x - 2y = 14$

3. $3y = 12x - 21$

Part K: Exponent Rules

Simplify. Write answers with positive exponents.

1. $5x^8 \cdot 6x^6$

9. $\frac{6x^9}{x^3}$

2. $5x^6 \cdot 8x^9$

10. $\frac{7x^8}{x^6}$

3. $2x^7 \cdot 2x^6$

11. $(6x^8)^3$

4. $2x^9 \cdot 2x^8$

12. $(5x^6)^4$

5. $4x^2 \cdot 3x^8$

13. $(4x^2)^4$

6. $\frac{4x^5}{x^4}$

14. $(7x^6)^3$

7. $\frac{3x^{11}}{x^3}$

15. $(7x^7)^2$

8. $\frac{2x^{12}}{x^4}$

16. $11 \cdot 5^{-3}$

17. $\frac{7}{7^{-1}}$

18. $\frac{10}{5^{-2}}$

25. $6^4 \cdot 6^{-2}$

19. $5 \cdot 6^{-2}$

26. $(2x^{-1})^3 \cdot x^7$

20. $4 \cdot (11^2)^{-2}$

27. $(10x^4y^3)^2 \cdot 3x^4y^4$

21. $10^4 \cdot 10^{-2}$

28. $\left(\frac{4x^{12}}{4x^8}\right)^2$

22. $5^3 \cdot 5^{-1}$

29. $\left(\frac{6x^8}{2x^6}\right)^3$

23. $3^6 \cdot 3^{-1}$

30. $(6x^7y^8)^3 \cdot 10x^5y^4$

24. $5^2 \cdot 5^{-5}$

Part L: Roots*Evaluate each root.*

1. $\sqrt{100}$

7. $\sqrt{121}$

2. $\sqrt{144}$

8. $\sqrt{4}$

3. $\sqrt{25}$

9. $\sqrt{81}$

4. $\sqrt{36}$

10. $\sqrt{49}$

5. $\sqrt{9}$

11. $\sqrt{16}$

6. $\sqrt{64}$

12. $\sqrt[3]{216}$

13. $\sqrt[3]{27}$

15. $\sqrt[3]{8}$

14. $\sqrt[3]{64}$

16. $\sqrt[3]{125}$

Part M: Simplifying Radicals*Simplify each radical completely.*

1. $\sqrt{40}$

8. $\sqrt{125}$

2. $\sqrt{12}$

9. $\sqrt[3]{144}$

3. $\sqrt{32}$

10. $\sqrt[3]{72}$

4. $\sqrt{104}$

11. $\sqrt[3]{80}$

5. $\sqrt{84}$

12. $\sqrt[3]{24}$

6. $\sqrt{116}$

13. $\sqrt[3]{32}$

7. $\sqrt{76}$

14. $\sqrt[3]{104}$

Answer Key

Part A: Negative Number Operations

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|------------|------------|------------|
| 1. -54 | 11. -81 | 21. 507 |
| 2. -819 | 12. 58 | 22. -27 |
| 3. 16 | 13. 183 | 23. 32 |
| 4. 100 | 14. 51 | 24. -26 |
| 5. -59 | 15. 11 | 25. 3 |
| 6. 2 | 16. 529 | 26. 27 |
| 7. -160 | 17. -284 | 27. 961 |
| 8. 18 | 18. 315 | 28. -384 |
| 9. -139 | 19. -15 | 29. 58 |
| 10. -121 | 20. 306 | 30. -8 |

Part B: Negative Number Operations

- | | | |
|-----------|------------|------------|
| 1. -4 | 11. 7 | 21. -11 |
| 2. -8 | 12. -132 | 22. 2 |
| 3. -11 | 13. 10 | 23. -18 |
| 4. -23 | 14. -132 | 24. -5 |
| 5. -9 | 15. -44 | 25. 93 |
| 6. 66 | 16. 13 | 26. -33 |
| 7. 10 | 17. 18 | 27. -21 |
| 8. -38 | 18. 63 | 28. 90 |
| 9. -15 | 19. -60 | 29. -147 |
| 10. -53 | 20. -14 | 30. -72 |

Part C: Solving Linear Equations

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|-----------------------|-----------------------|-------------------------|
| 1. $x = -5$ | 11. $x = -1$ | 21. $x = 14$ |
| 2. $x = -3$ | 12. $x = -3$ | 22. $x = 4$ |
| 3. $x = 6$ | 13. $x = -7$ | 23. $x = -\frac{3}{2}$ |
| 4. $x = 10$ | 14. $x = -3$ | 24. $x = 2$ |
| 5. $x = \frac{7}{2}$ | 15. $x = -7$ | 25. $x = \frac{18}{5}$ |
| 6. $x = 2$ | 16. $x = -10$ | 26. $x = -15$ |
| 7. $x = 2$ | 17. $x = \frac{4}{3}$ | 27. $x = 19$ |
| 8. $x = -5$ | 18. $x = 4$ | 28. $x = 3$ |
| 9. $x = -\frac{3}{2}$ | 19. $x = 15$ | 29. $x = -14$ |
| 10. $x = 10$ | 20. $x = -14$ | 30. $x = -\frac{10}{3}$ |

Part D: Linear Inequalities

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|----------------|----------------|------------------|
| 1. $x > -6$ | 6. $x \leq 10$ | 11. $x > -3$ |
| 2. $x < -14$ | 7. $x > -4$ | 12. $x \leq -12$ |
| 3. $x \geq 13$ | 8. $x > 2$ | 13. $x \leq 15$ |
| 4. $x \geq -7$ | 9. $x > 12$ | 14. $x > 4$ |
| 5. $x < 5$ | 10. $x \leq 8$ | 15. $x < 16$ |

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|-----------------|------------------|----------------|
| 16. $x > 4$ | 21. $x < 5$ | 26. $x \leq 8$ |
| 17. $x < 7$ | 22. $x \geq -5$ | 27. $x > -6$ |
| 18. $x < -12$ | 23. $x \leq -20$ | 28. $x < 3$ |
| 19. $x \geq 5$ | 24. $x > -3$ | 29. $x < -4$ |
| 20. $x \leq 14$ | 25. $x < 7$ | 30. $x < 8$ |

Part E: Slope and Linear Graphs

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|-----------------------------------|--------------------------|-------------------------|
| 1. $m = \frac{7}{2}, \quad b = 6$ | 3. $m = 8, \quad b = 7$ | 5. $m = 9, \quad b = 1$ |
| 2. $m = 6, \quad b = 9$ | 4. $m = 5, \quad b = -6$ | |

Part F: Slope Through Two Points

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|------------|----------------------|----------------------|
| 1. $m = 2$ | 3. $m = \frac{5}{2}$ | 5. $m = \frac{1}{2}$ |
| 2. $m = 3$ | 4. $m = 3$ | |

Part G: Equations From Two Points

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|---------------------------|---------------------------|---------------------------|
| 1. $y = \frac{1}{2}x + 2$ | 3. $y = -6$ | 5. $y = \frac{5}{2}x + 2$ |
| 2. $y = \frac{7}{2}x + 7$ | 4. $y = \frac{3}{2}x + 3$ | |

Part H: Equations From a Slope and a Point

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|---------------------------|---------------------------|---------------------------|
| 1. $y = \frac{4}{3}x + 2$ | 3. $y = \frac{9}{5}x + 3$ | 5. $y = \frac{5}{3}x + 5$ |
| 2. $y = \frac{8}{3}x$ | 4. $y = \frac{1}{2}x$ | |

Part I: Slope and Equation From Two Points

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|---|--|--|
| 1. $m = \frac{11}{3}, \quad y = \frac{11}{3}x + 10$ | 3. $m = 3, \quad y = 3x + 2$ | 5. $m = \frac{5}{2}, \quad y = \frac{5}{2}x - 9$ |
| 2. $m = 0, \quad y = -7$ | 4. $m = \frac{9}{4}, \quad y = \frac{9}{4}x + 5$ | |

Part J: Rewriting in Slope-Intercept Form

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|------------------------------------|--------------------------|--------------------------|
| 1. $m = \frac{7}{2}, \quad b = -1$ | 3. $m = 4, \quad b = -7$ | 5. $m = 3, \quad b = -7$ |
| 2. $m = 2, \quad b = 2$ | 4. $m = 4, \quad b = 3$ | |

Part K: Exponent Rules

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|---------------|-----------------------|------------------------|
| 1. $30x^{14}$ | 11. $216x^{24}$ | 21. 100 |
| 2. $40x^{15}$ | 12. $625x^{24}$ | 22. 25 |
| 3. $4x^{13}$ | 13. $256x^8$ | 23. 243 |
| 4. $4x^{17}$ | 14. $343x^{18}$ | 24. $\frac{1}{125}$ |
| 5. $12x^{10}$ | 15. $49x^{14}$ | 25. 36 |
| 6. $4x$ | 16. $\frac{11}{125}$ | 26. $8x^4$ |
| 7. $3x^8$ | 17. 49 | 27. $300x^{12}y^{10}$ |
| 8. $2x^8$ | 18. 250 | 28. x^8 |
| 9. $6x^6$ | 19. $\frac{5}{36}$ | 29. $27x^6$ |
| 10. $7x^2$ | 20. $\frac{4}{14641}$ | 30. $2160x^{26}y^{28}$ |

Part L: Roots

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|-------|-------|-------|
| 1. 10 | 7. 11 | 13. 3 |
| 2. 12 | 8. 2 | 14. 4 |
| 3. 5 | 9. 9 | 15. 2 |
| 4. 6 | 10. 7 | 16. 5 |
| 5. 3 | 11. 4 | |
| 6. 8 | 12. 6 | |

Part M: Simplifying Radicals

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|-----------------|--------------------|---------------------|
| 1. $2\sqrt{10}$ | 6. $2\sqrt{29}$ | 11. $2\sqrt[3]{10}$ |
| 2. $2\sqrt{3}$ | 7. $2\sqrt{19}$ | 12. $2\sqrt[3]{3}$ |
| 3. $4\sqrt{2}$ | 8. $5\sqrt{5}$ | 13. $2\sqrt[3]{4}$ |
| 4. $2\sqrt{26}$ | 9. $2\sqrt[3]{18}$ | 14. $2\sqrt[3]{13}$ |
| 5. $2\sqrt{21}$ | 10. $2\sqrt[3]{9}$ | |